Advanced Causal Inference

Lecture 17: Dependent Data/Interference

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This lecture note is partially based on the lecture notes/textbook of Ying Zhou

Outline

1. Overview
2. Examples
3. Interference
4. Partial Interference
5. Cluster Randomized Trial
6. Network Interference
7. Additional Literature
8. Assignment

Overview

- We will consider causal inference methods under dependent settings
  - clustered data
  - general network data

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Stable Unit Treatment Value Assumption (SUTVA)

Assumption (SUTVA (Rubin, 1980))

- *No interference between units (Cox, 1958)*
- *No different versions of a treatment; also known as (causal) consistency (Cole and Frangakis, 2009).*

1. No interference between units
   - Interference: the potential outcome $Y_i(a)$ for an individual i depends on what treatment other people receive
   - Examples: vaccination, advertising, infectious diseases, social networks, agricultural experiments
   - Later lecture on violation of this assumption

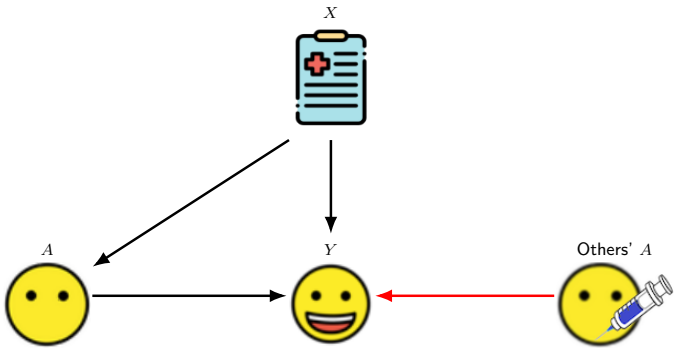
2. Consistency: No different versions of a treatment
   - Sometimes, a treatment A doesn't have a clear meaning, as it has many versions
   - Example: Different surgeons performing the same procedure
   - Rule out the case that potential outcomes depend on the specific version of the treatment received

- Under SUTVA, each unit has only two potential outcomes: $Y_i(1)$ and $Y_i(0)$.
- $Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0) = Y_i(A_i) \quad \Leftrightarrow \quad$ if $A_i = 1$ then $Y_i = Y_i(1)$; if $A_i = 0$ then $Y_i = Y_i(0)$
- SUTVA connects the intervention we see (A) with the causal intervention of interest $(A = a)$

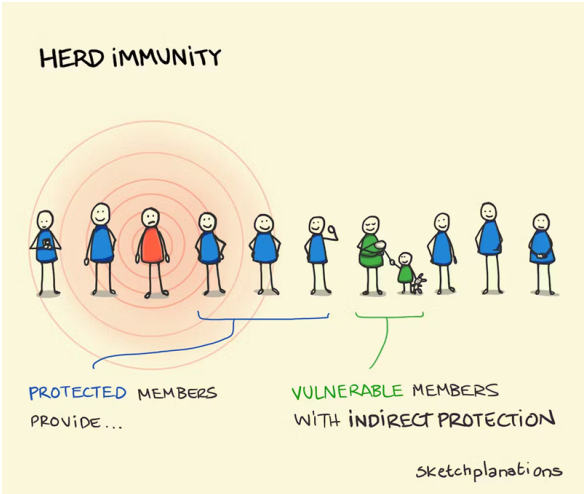
Violation of SUTVA: Infectious Disease

- A = COVID-19 vaccine status, Y = infection status

- Y depends not only on his/her own A, but also on others' A

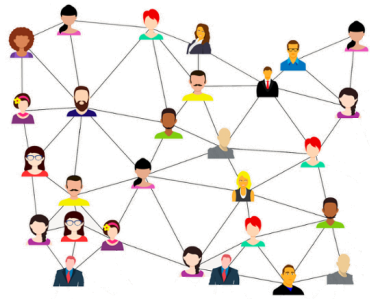


Violation of SUTVA: Herd Immunity



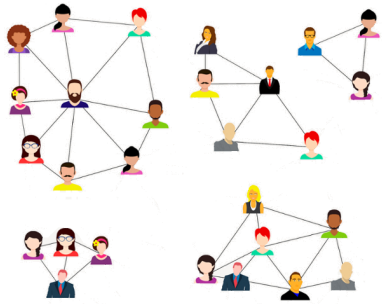
Violation of SUTVA: Cai et al. (2015)

- Randomized experiment in rural China
- A: An informational session on the benefits of weather insurance
- Y: Whether a farmer purchases weather insurance
- Individuals may share details about the insurance with their neighbors



Violation of SUTVA: Barrera-Osorio et al. (2011)

- Randomized experiments in Colombia
- A: Conditional Cash Transfer (CCT) program
  $A = 1$: students are eligible to receive cash if they meet certain academic requirements
- Y: School attendance rate
- Receiving a CCT increases household net income, which may influence the school attendance rates of siblings
- However, receiving a CCT is unlikely to affect students in other households



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Interference (Narrowly Defined)

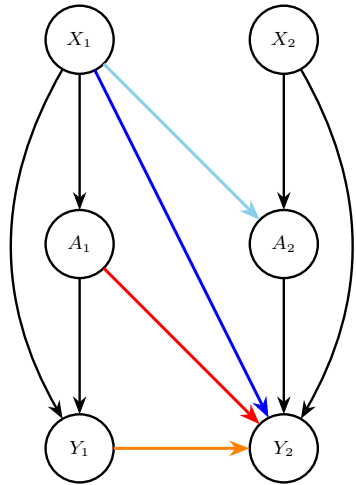
- Let N denote the total number of units in the population
- Let $\boldsymbol{a} = (a_1, \ldots, a_N)^\top \in \{0,1\}^N$ represent a possible treatment assignment vector for the N units
- Let $Y_i(\boldsymbol{a}) = Y_i(a_1, \ldots, a_N)$ denote the potential outcome of unit i if the treatment assignment of N units were $\boldsymbol{a}$
  - Under SUTVA, we assume

$$Y_i(\boldsymbol{a}) = Y_i(a_i) \,, \quad \forall i = 1, \ldots, N \tag{1}$$

  - Unit i's potential outcome depends only on its own treatment assignment
- **Interference** (Cox, 1958): Condition (1) does not hold
  - The potential outcome of some units depends on the treatment assignments of other units

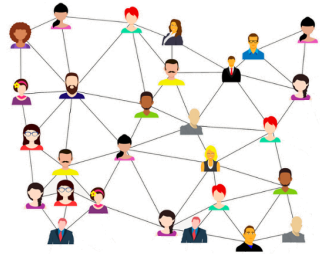
Interference (Broadly Defined)

- Interference: →
  - One's A affects others' Y
- Covariate Interference: →
  - One's X affects others' Y
- Covariate-Treatment Interference: →
  - One's X affects others' Y
- Outcome Interference/Contagion: →
  - One's Y affects others' Y



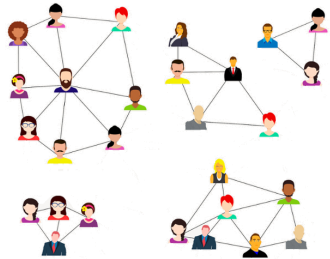
Network v. Partial Interference

- Network/general interference
  interference may arise across nonseparable, large network



- Partial/clustered interference (Sobel, 2006; Hudgens and Halloran, 2008)
  interference only arises within non-overlapping clusters

- $Y_i(\boldsymbol{a}) = Y_i(\boldsymbol{a}_{\mathcal{N}_i})$ where $\mathcal{N}_i = \{j \,|\, i \text{ and } j \text{ belong to the same cluster}\}$



Complications Arising from Interference

- Under no interference, only one treatment effect can be defined for each individual:

$$\tau_i = Y_i(1) - Y_i(0)$$

- But, under interference, "many" treatment effects can be defined:

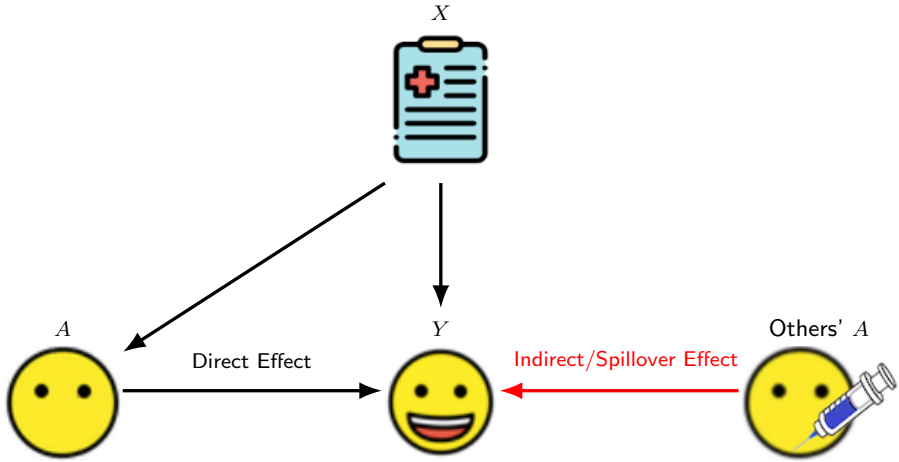
$Y_i(a_i = 1, \boldsymbol{a}_{(-i)} = \mathbf{0}) - Y_i(a_i = 0, \boldsymbol{a}_{(-i)} = \mathbf{0})$: effect of receiving treatment when all other units do not receive treatment

$Y_i(a_i = 1, \boldsymbol{a}_{(-i)} = \mathbf{1}) - Y_i(a_i = 0, \boldsymbol{a}_{(-i)} = \mathbf{1})$: effect of receiving treatment when all other units receive treatment

$Y_i(a_i = 0, \boldsymbol{a}_{(-i)} = \mathbf{1}) - Y_i(a_i = 0, \boldsymbol{a}_{(-i)} = \mathbf{0})$: effect of changing the treatment status of other units when unit i does not receive treatment

$Y_i(a_i = 1, \boldsymbol{a}_{(-i)} = \mathbf{1}) - Y_i(a_i = 1, \boldsymbol{a}_{(-i)} = \mathbf{0})$: effect of changing the treatment status of other units when unit i receives treatment

(Individual-level) Direct and Spillover Effects



$Y_i(a_i = 1, \boldsymbol{a}_{(-i)}) - Y_i(a_i = 0, \boldsymbol{a}_{(-i)})$: direct effect

$Y_i(a_i, \boldsymbol{a}_{(-i)}) - Y_i(a_i, \boldsymbol{a}'_{(-i)})$: indirect/spillover effect

Exposure Mapping

- There are 2^N potential outcomes for each unit i, so the number of possible treatment effects is $\binom{2^N}{2}$
- **Exposure mapping** (Aronow and Samii, 2017):

  For unit i, there exists a function $d_i : \{0,1\}^{\otimes N} \to \Delta \in \mathbb{R}^k$ such that

$$Y_i(\boldsymbol{a}) = Y_i(d_i(\boldsymbol{a})) \ , \quad \forall \boldsymbol{a} \in \{0,1\}^{\otimes N}$$

- **Exposure**: $d_i(\boldsymbol{a}) \in \Delta^k$
  - $d_i(\boldsymbol{a}) = d_i(\boldsymbol{a}')$ $\Rightarrow$ $\boldsymbol{a}$ and $\boldsymbol{a}'$ are effectively identical for unit i's potential outcome
- Examples
  - $d_i(\boldsymbol{a}) = \sum_j a_j$; $Y_i(\boldsymbol{a}) = Y_i(\sum_j a_j)$
    depends only on the total number of treated units
  - $d_i(\boldsymbol{a}) = (a_i, \sum_{j \neq i} a_j)$; $Y_i(\boldsymbol{a}) = Y_i(a_i, \sum_{j \neq i} a_j)$
    depends on its own treatment status and the total number of treated units among the remaining units
  - $d_i(\boldsymbol{a}) = (\boldsymbol{a}_{\mathcal{N}_i})$; $Y_i(\boldsymbol{a}) = Y_i(\boldsymbol{a}_{\mathcal{N}_i})$
    depends only on the treatment status of units within the same cluster

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Estimands Under Partial Interference

- Seminal papers: Sobel (2006); Hudgens and Halloran (2008); Tchetgen Tchetgen and VanderWeele (2012)
- Use double subscript ij to denote cluster $i \in \{1, \ldots, M\}$ unit $j \in \{1, \ldots, N_i\}$
- $\boldsymbol{V}_i = (V_{i1}, \ldots, V_{iN_i})^\intercal$ = vector of V for cluster i
- $\boldsymbol{V}_{i(-j)}$ = vector of V for cluster i, excluding unit j
- $Y_{ij}(a_{ij}, \boldsymbol{a}_{i(-j)})$ = potential outcome of unit ij if $A_{ij} = a_{ij}$ and $\boldsymbol{A}_{i(-j)} = \boldsymbol{a}_{i(-j)}$

Estimands Under Partial Interference

- α-**Bernoulli Policy**:
  - Consider a future treatment assignment where $A_{ij} \overset{iid}{\sim} \text{Ber}(\alpha)$
  - Each unit receives treatment with prob α independently of others' treatment status
  - Note that the observed data may not have been generated under this policy
- $\pi(\cdot; \alpha)$ = probability of trt allocation under α-B policy

$$\pi(\boldsymbol{a}_i; \alpha) = \prod_{\ell=1}^{N_i} \alpha^{a_{i\ell}} (1-\alpha)^{1-a_{i\ell}}$$

$$\pi(\boldsymbol{a}_{i(-j)}; \alpha) = \prod_{\ell \neq j} \alpha^{a_{i\ell}} (1-\alpha)^{1-a_{i\ell}}$$

Estimands Under Partial Interference

- The expected individual-level potential outcomes under α-B policy are

$$\mathcal{Y}_{ij}(\alpha) = \sum_{\boldsymbol{a}_i} Y_{ij}(a_{ij}, \boldsymbol{a}_{i(-j)})\pi(\boldsymbol{a}_i; \alpha) \qquad \mathcal{Y}_{ij}(a; \alpha) = \sum_{\boldsymbol{a}_{i(-j)}} Y_{ij}(a, \boldsymbol{a}_{i(-j)})\pi(\boldsymbol{a}_{i(-j)}; \alpha)$$

- Aggregate at the cluster-level:

$$\overline{\mathcal{Y}}_i(\alpha) = \frac{1}{N_i}\sum_{j=1}^{N_i} \mathcal{Y}_{ij}(\alpha) \qquad \overline{\mathcal{Y}}_i(a; \alpha) = \frac{1}{N_i}\sum_{j=1}^{N_i} \mathcal{Y}_{ij}(a; \alpha)$$

- Take the expectation (or average) over M clusters:
  - Finite-population framework:

$$\mu(\alpha) = \frac{1}{M}\sum_{i=1}^{M} \overline{\mathcal{Y}}_i(\alpha) \qquad \mu(a; \alpha) = \frac{1}{M}\sum_{i=1}^{M} \overline{\mathcal{Y}}_i(a; \alpha)$$

  - Infinite-population framework:

$$\mu(\alpha) = \mathrm{E}\big[\overline{\mathcal{Y}}_i(\alpha)\big] \qquad \mu(a; \alpha) = \mathrm{E}\big[\overline{\mathcal{Y}}_i(a; \alpha)\big]$$

Estimands Under Partial Interference

- Overall effect: $\mathsf{OE}(\alpha', \alpha) = \mu(\alpha') - \mu(\alpha)$
  - Change in the expected outcome between α' and α-policies
- Direct effect: $\mathsf{DE}(\alpha) = \mu(1; \alpha) - \mu(0; \alpha)$
  - Change in the expected ego outcome between trt and ctrl, given that neighbors receive trt with prob α
- Indirect (spillover) effect: $\mathsf{IE}(\alpha', \alpha) = \mu(0; \alpha') - \mu(0; \alpha)$
  - Change in the expected ego outcome when neighbors receive trt with prob α' vs. α, given that the ego is untreated
  - One can define another spillover effect: $\mathsf{IE}(\alpha', \alpha) = \mu(1; \alpha') - \mu(1; \alpha)$
- Total effect: $\mathsf{TE}(\alpha', \alpha) = \mu(1; \alpha') - \mu(0; \alpha)$
  - Change in the expected ego outcome when the ego receives trt vs. ctrl and neighbors receive trt with prob α' vs. α
  - $\mathsf{TE}(\alpha', \alpha) = \mathsf{DE}(\alpha') + \mathsf{IE}(\alpha', \alpha)$

Identification under Partial Interference

- Consider infinite-population

Assumption (Consistency)

$Y_{ij} = Y_{ij}(\boldsymbol{A}_i)$

Assumption (Conditional Ignorability)

$Y_{ij}(\boldsymbol{a}_i) \perp\!\!\!\perp \boldsymbol{A}_i \,|\, \boldsymbol{X}_i,\ \forall \boldsymbol{a}_i$

Assumption (Positivity)

$\Pr(\boldsymbol{A}_i = \boldsymbol{a}_i | \boldsymbol{X}_i) > 0$ *for all* $\boldsymbol{a}_i$

- $\mu(\alpha)$ and $\mu(a;\alpha)$ are identified:

$$\mu(\alpha) = \mathrm{E}\left[\frac{1}{N_i}\sum_{j=1}^{N_i}\frac{Y_{ij}\pi(\boldsymbol{A}_i;\alpha)}{\Pr(\boldsymbol{A}_i \,|\, \boldsymbol{X}_i)}\right] = \mathrm{E}\left[\frac{1}{N_i}\sum_{j=1}^{N_i}\sum_{\boldsymbol{a}_i}\mathrm{E}\big[Y_{ij}|\boldsymbol{A}_i = \boldsymbol{a}_i, \boldsymbol{X}_i\big]\pi(\boldsymbol{a}_i;\alpha)\right]$$

$$\mu(a,\alpha) = \mathrm{E}\left[\frac{1}{N_i}\sum_{j=1}^{N_i}\frac{\mathbb{1}(A_{ij} = a)Y_{ij}\pi(\boldsymbol{A}_{i(-j)};\alpha)}{\Pr(\boldsymbol{A}_i \,|\, \boldsymbol{X}_i)}\right] = \mathrm{E}\left[\frac{1}{N_i}\sum_{j=1}^{N_i}\sum_{\boldsymbol{a}_i}\mathrm{E}\big[Y_{ij}|A_{ij} = a, \boldsymbol{A}_{i(-j)} = \boldsymbol{a}_{i(-j)}, \boldsymbol{X}_i\big]\pi(\boldsymbol{a}_{i(-j)};\alpha)\right]$$

Estimation under Partial Interference

- Identification: using $e(\boldsymbol{A}_i|\boldsymbol{X}_i) = \Pr(\boldsymbol{A}_i \mid \boldsymbol{X}_i)$ and $m_{ij}(a_{ij}, \boldsymbol{a}_{i(-j)}, \boldsymbol{X}_i) = \mathrm{E}\big[Y_{ij}|A_{ij} = a, \boldsymbol{A}_{i(-j)} = \boldsymbol{a}_{i(-j)}, \boldsymbol{X}_i\big]$

$$\mu(\alpha) = \mathrm{E}\left[\frac{1}{N_i}\sum_{j=1}^{N_i}\frac{Y_{ij}\pi(\boldsymbol{A}_i;\alpha)}{e(\boldsymbol{A}_i \mid \boldsymbol{X}_i)}\right] = \mathrm{E}\left[\frac{1}{N_i}\sum_{j=1}^{N_i}\sum_{\boldsymbol{a}_i} m_{ij}(a_{ij}, \boldsymbol{a}_{i(-j)}, \boldsymbol{X}_i)\pi(\boldsymbol{a}_i;\alpha)\right]$$

$$\mu(a,\alpha) = \mathrm{E}\left[\frac{1}{N_i}\sum_{j=1}^{N_i}\frac{\mathbb{1}(A_{ij} = a)Y_{ij}\pi(\boldsymbol{A}_{i(-j)};\alpha)}{e(\boldsymbol{A}_i \mid \boldsymbol{X}_i)}\right] = \mathrm{E}\left[\frac{1}{N_i}\sum_{j=1}^{N_i}\sum_{\boldsymbol{a}_i} m_{ij}(a, \boldsymbol{a}_{i(-j)}, \boldsymbol{X}_i)\pi(\boldsymbol{a}_{i(-j)};\alpha)\right]$$

- We need to estimate e and m_{ij}, but this task is very challenging
  - The dimension of the regressors varies across clusters; this does not happen in standard regression models
  - The joint propensity score has 2^{N_i} levels, whereas a standard logistic regression has only two levels
  - m_{ij} can be unit-specific (i.e., each unit may have a different functional form)
    this issue is more pronounced when units within a cluster have an ordering and are therefore not exchangeable.

- Need additional assumptions:
  - Parametric models: Tchetgen Tchetgen and VanderWeele (2012); Perez-Heydrich et al. (2014); Liu et al. (2016, 2019); Barkley et al. (2020); Kang et al. (2023)
  - Structural assumptions with "simple" exposure mapping: Park and Kang (2022); Park et al. (2024)

Estimation under Partial Interference

- The three commonly used estimators discussed in the literature are:

Hudgens and Halloran (2008); Tchetgen Tchetgen and VanderWeele (2012); Perez-Heydrich et al. (2014); Liu et al. (2016, 2019); Barkley et al. (2020); Park and Kang (2022)

$$\widehat{\mu}_{\mathsf{IPW}}(\alpha) = \frac{1}{M}\sum_{i=1}^{M}\frac{1}{N_i}\sum_{j=1}^{N_i}\frac{Y_{ij}\pi(\boldsymbol{A}_i;\alpha)}{\widehat{e}(\boldsymbol{A}_i \,|\, \boldsymbol{X}_i)}$$

$$\widehat{\mu}_{\mathsf{OR}}(\alpha) = \frac{1}{M}\sum_{i=1}^{M}\frac{1}{N_i}\sum_{j=1}^{N_i}\sum_{\boldsymbol{a}_i}\widehat{m}_{ij}(a_{ij},\boldsymbol{a}_{i(-j)},\boldsymbol{X}_i)\pi(\boldsymbol{a}_i;\alpha)$$

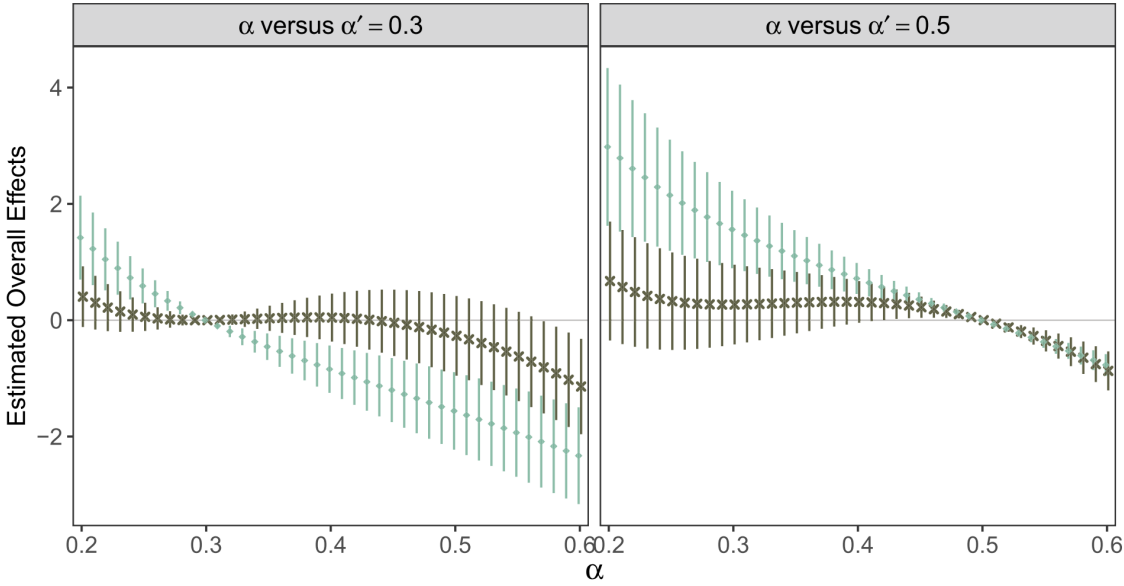
$$\widehat{\mu}_{\mathsf{DR}}(\alpha) = \frac{1}{M}\sum_{i=1}^{M}\frac{1}{N_i}\sum_{j=1}^{N_i}\sum_{\boldsymbol{a}_i}\left\{\frac{\mathbb{1}(\boldsymbol{A}_i=\boldsymbol{a}_i)Y_{ij}-\widehat{m}_{ij}(a_{ij},\boldsymbol{a}_{i(-j)},\boldsymbol{X}_i)}{\widehat{e}(\boldsymbol{a}_i \,|\, \boldsymbol{X}_i)}+\widehat{m}_{ij}(a_{ij},\boldsymbol{a}_{i(-j)},\boldsymbol{X}_i)\right\}\pi(\boldsymbol{a}_i;\alpha)$$

- Park and Kang (2022): the DR estimator by Liu et al. (2019) is semiparametric efficient in a certain model

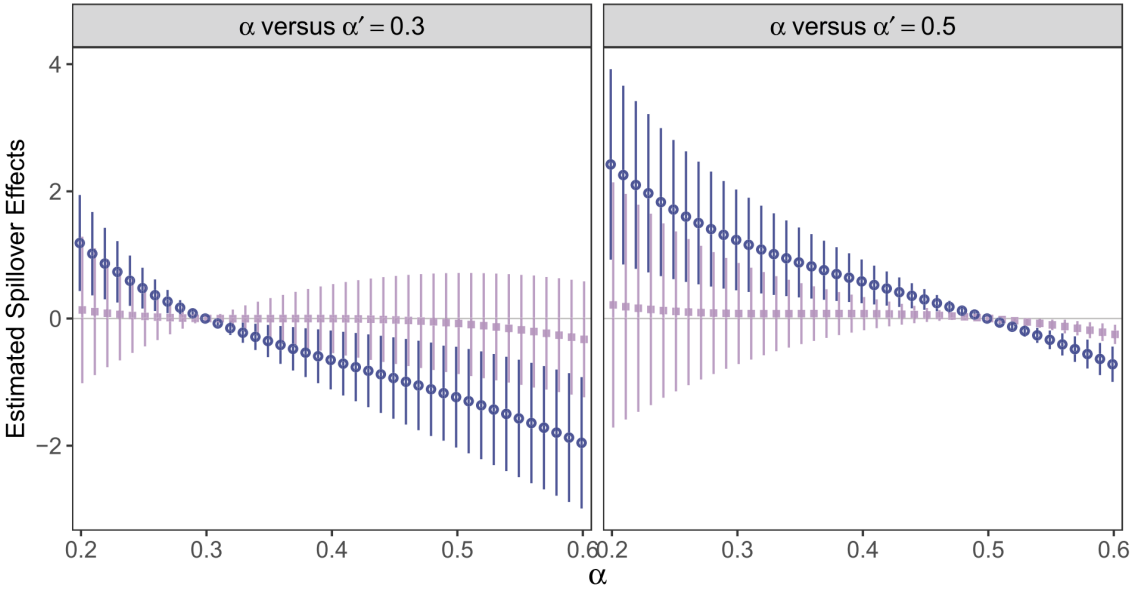
- Bootstrap should be performed at the cluster level;
  i.e., entire clusters are sampled, with all their units included or excluded together

Example: Analysis of Cholera Vaccine Study in Matlab, Bangladesh

- Barkley et al. (2020)
- Units: 121975 people from 6415 villages
- Treatment: Receive cholera vaccine or none
- Outcome: Cholera infection
- Partial interference: transmission of cholera often takes place within villages



Overall effect; Black = Tchetgen Tchetgen and VanderWeele (2012); Green = Barkley et al. (2020)



Spillover effect; Blue $= \mu(0;\alpha') - \mu(0;\alpha)$; Lavender $= \mu(1;\alpha') - \mu(1;\alpha)$

Example: Simulated Data

```
expit <- function(v){
  exp(v)/(1+exp(v))
}
M <- 1000; N <- 2
X <- rnorm(M*N); A <- rbinom(M*N,1,expit(0.25*X))
Y11 <- 1.5+X + 0.25*rnorm(M*N) # Y(ego=1,peer=1)
Y01 <- 0.5+X + 0.25*rnorm(M*N) # Y(ego=0,peer=1)
Y10 <- 1+X + 0.25*rnorm(M*N) # Y(ego=1,peer=0)
Y00 <- X + 0.25*rnorm(M*N) # Y(ego=0,peer=0)
```

$$X_{ij} \sim N(0,1); \quad A_{ij} \sim \mathsf{Ber}(\mathsf{expit}(0.25X_{ij}))$$

$$Y_{ij}(a_{ij}, a_{i(-j)}) = a_{ij} + 0.5a_{i(-j)}X_{ij} + 0.25\epsilon_{ij}$$

$$\mathsf{OE}(\alpha', \alpha) = 1.5(\alpha' - \alpha)$$

$$\mathsf{DE}(\alpha) = 1$$

$$\mathsf{IE}(\alpha', \alpha) = 0.5(\alpha' - \alpha)$$

```
pos.unit1 <- seq(1,M*N,by=2); pos.unit2 <- seq(2,M*N,by=2)
A.unit1 <- A[pos.unit1]; A.unit2 <- A[pos.unit2]; X.unit1 <- X[pos.unit1]; X.unit2 <- X[pos.unit2]

Y.unit1 <- Y11[pos.unit1]*A.unit1*A.unit2 + Y01[pos.unit1]*(1-A.unit1)*A.unit2 +
  Y10[pos.unit1]*A.unit1*(1-A.unit2) + Y00[pos.unit1]*(1-A.unit1)*(1-A.unit2)

Y.unit2 <- Y11[pos.unit2]*A.unit2*A.unit1 + Y01[pos.unit2]*(1-A.unit2)*A.unit1 +
  Y10[pos.unit2]*A.unit2*(1-A.unit1) + Y00[pos.unit2]*(1-A.unit2)*(1-A.unit1)

Y <- as.vector(rbind(Y.unit1,Y.unit2))
A.peer <- as.vector(rbind(A.unit2,A.unit1))
X.peer <- as.vector(rbind(X.unit2,X.unit1))

Data <- data.frame(Y=Y,A.ego=A,A.peer=A.peer,X.ego=X,X.peer=X.peer)
```

Example: Simulated Data

```
policy <- function(alpha,A1,A2){ alpha^(A1+A2)*(1-alpha)^(2-A1-A2) }
policy.peer <- function(alpha,A){ alpha^(A)*(1-alpha)^(1-A) }

## Propensity: Assuming Pr(A1=a1,A2=a2|X1,X2) = Pr(A1=a1|X1,X2)*Pr(A2=a2|X2,X1)
PS.Est   <- glm(A.ego~X.ego+X.peer,data=Data,family="binomial")
PS.Fit   <- predict(PS.Est,type="response")
PS.unit1 <- PS.Fit[pos.unit1]; PS.unit2 <- PS.Fit[pos.unit2]
CPS.Est  <- (PS.unit1*A.unit1+(1-PS.unit1)*(1-A.unit1))*
  (PS.unit2*A.unit2+(1-PS.unit2)*(1-A.unit2))

## IPW
mu.IPW <- function(alpha){
  mean( ((policy(alpha,A.unit1,A.unit2)*Y.unit1 +
            policy(alpha,A.unit2,A.unit1)*Y.unit2))/2/CPS.Est )
}
mu.a.IPW <- function(a,alpha){
  mean( (policy.peer(alpha,A.unit2)*as.numeric(A.unit1==a)*Y.unit1 +
           policy.peer(alpha,A.unit1)*as.numeric(A.unit2==a)*Y.unit2)/2/CPS.Est )
}
```

$$\pi(A_{i1}, A_{i2}; \alpha) = \alpha^{A_{i1}+A_{i2}}(1-\alpha)^{2-A_{i1}-A_{i2}}$$

$$\pi(A_{ij}; \alpha) = \alpha^{A_{ij}}(1-\alpha)^{1-A_{ij}}$$

$$\Pr(\boldsymbol{A}_i|\boldsymbol{X}_i) = \Pr(A_{i1}|X_{i1}, X_{i2}) \Pr(A_{i2}|X_{i2}, X_{i1})$$

$$\text{logit}\Pr(A_{ij} = 1|X_{ij}, X_{i(-j)}) = \beta_0 + \beta_1 X_{ij} + \beta_2 X_{i(-j)}$$

$$\hat{\mu}_{\text{IPW}}(\alpha) = \frac{1}{M}\sum_{i=1}^{M}\frac{1}{N_i}\sum_{j=1}^{N_i}\frac{Y_{ij}\pi(\boldsymbol{A}_i;\alpha)}{\hat{e}(\boldsymbol{A}_i \,|\, \boldsymbol{X}_i)}$$

$$\hat{\mu}_{\text{IPW}}(a,\alpha) = \frac{1}{M}\sum_{i=1}^{M}\frac{1}{N_i}\sum_{j=1}^{N_i}\frac{\mathbb{1}(A_{ij}=a)Y_{ij}\pi(A_{i(-j)};\alpha)}{\hat{e}(\boldsymbol{A}_i \,|\, \boldsymbol{X}_i)}$$

Example: Simulated Data

```
## OR
OR.Est <- lm(Y~A.ego+A.peer+X.ego+X.peer, data=Data)
OR.Fit <- function(alpha,a1,a2,X1,X2){
  predict(OR.Est, newdata=data.frame(A.ego=a1, A.peer=a2, X.ego=X1, X.peer=X2))
}
mu.OR <- function(alpha){
  mean( policy(alpha,1,1)*(OR.Fit(alpha,1,1,X.unit1,X.unit2) +
                             OR.Fit(alpha,1,1,X.unit2,X.unit1))/2 +
          policy(alpha,0,1)*(OR.Fit(alpha,0,1,X.unit1,X.unit2) +
                               OR.Fit(alpha,0,1,X.unit2,X.unit1))/2 +
          policy(alpha,1,0)*(OR.Fit(alpha,1,0,X.unit1,X.unit2) +
                               OR.Fit(alpha,1,0,X.unit2,X.unit1))/2 +
          policy(alpha,0,0)*(OR.Fit(alpha,0,0,X.unit1,X.unit2) +
                               OR.Fit(alpha,0,0,X.unit2,X.unit1))/2)
}
mu.a.OR <- function(a,alpha){
  mean( policy.peer(alpha,1)*(OR.Fit(alpha,a,1,X.unit1,X.unit2) +
                                OR.Fit(alpha,a,1,X.unit2,X.unit1))/2 +
          policy.peer(alpha,0)*(OR.Fit(alpha,a,0,X.unit1,X.unit2) +
                                  OR.Fit(alpha,a,0,X.unit2,X.unit1))/2)
}
```

$$
\begin{aligned}
&\mathrm{E}(Y_{ij}|\boldsymbol{A}_i, \boldsymbol{X}_i) \\
&= m(A_{ij}, A_{i(-j)}, X_{ij}, X_{i(-j)}) \\
&= \gamma_0 + \gamma_1 A_{ij} + \gamma_2 A_{i(-j)} + \gamma_3 X_{ij} + \gamma_4 X_{i(-j)}
\end{aligned}
$$

$$
\begin{aligned}
&\widehat{\mu}_{\mathsf{OR}}(\alpha) \\
&= \frac{1}{M}\sum_{i=1}^{M}\frac{1}{N_i}\sum_{j=1}^{N_i}\sum_{\boldsymbol{a}_i}\widehat{m}(a_{ij}, a_{i(-j)}, X_{ij}, X_{i(-j)})\pi(\boldsymbol{a}_i;\alpha)
\end{aligned}
$$

$$
\begin{aligned}
&\widehat{\mu}_{\mathsf{OR}}(a,\alpha) \\
&= \frac{1}{M}\sum_{i=1}^{M}\frac{1}{N_i}\sum_{j=1}^{N_i}\sum_{a'=0}^{1}\widehat{m}(a, a', X_{ij}, X_{i(-j)})\pi(a';\alpha)
\end{aligned}
$$

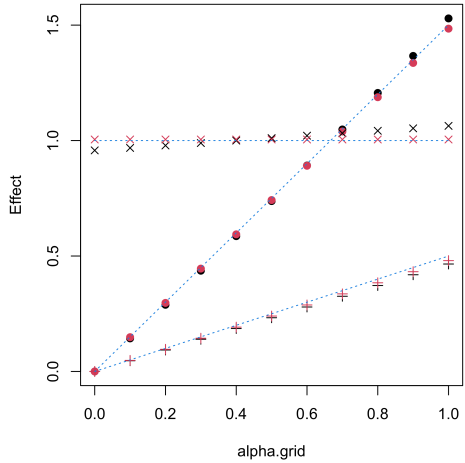
Example: Simulated Data

```
alpha.grid <- seq(0,1,by=0.1)

plot(alpha.grid,
     sapply(alpha.grid,function(v){mu.IPW(v)}-mu.IPW(0)),pch=19)
points(alpha.grid,
       sapply(alpha.grid,function(v){mu.OR(v)}-mu.OR(0)),col=2,pch=19)

points(alpha.grid,
       sapply(alpha.grid,function(v){mu.a.IPW(1,v)-mu.a.IPW(0,v)}),pch=4)
points(alpha.grid,
       sapply(alpha.grid,function(v){mu.a.OR(1,v)-mu.a.OR(0,v)}),col=2,pch=4)

points(alpha.grid,
       sapply(alpha.grid,function(v){mu.a.IPW(0,v)}-mu.a.IPW(0,0)),pch=3)
points(alpha.grid,
       sapply(alpha.grid,function(v){mu.a.OR(0,v)}-mu.a.OR(0,0)),col=2,pch=3)
```



Black = IPW; Red = OR

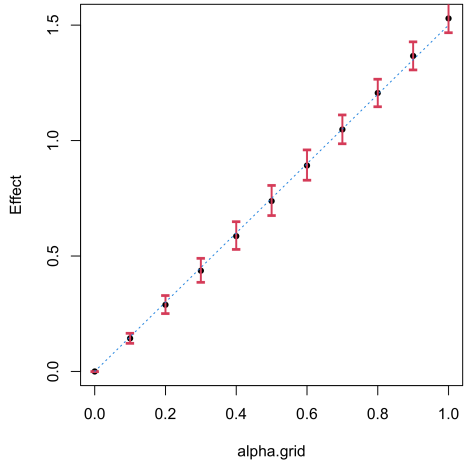
- $\widehat{\mathsf{OE}}(\alpha', \alpha = 0) \simeq 1.5\alpha'$
- $\widehat{\mathsf{DE}}(\alpha) \simeq 1$
- $\widehat{\mathsf{IE}}(\alpha', \alpha = 0) \simeq 0.5\alpha'$

Example: Simulated Data

```
OE <- sapply(alpha.grid,function(v){mu.IPW(v)}-mu.IPW(0))

OE.IPW.Boot <- function(Data.Boot){
  A.unit1 <- Data.Boot$A.ego[pos.unit1]; A.unit2 <- Data.Boot$A.ego[pos.unit2]
  Y.unit1 <- Data.Boot$Y[pos.unit1]; Y.unit2 <- Data.Boot$Y[pos.unit2]
  PS.Est   <- glm(A.ego~X.ego+X.peer,data=Data.Boot,family="binomial")
  PS.Fit   <- predict(PS.Est,type="response")
  PS.unit1 <- PS.Fit[pos.unit1]; PS.unit2 <- PS.Fit[pos.unit2]
  CPS.Est  <- (PS.unit1*A.unit1+(1-PS.unit1)*(1-A.unit1))*
    (PS.unit2*A.unit2+(1-PS.unit2)*(1-A.unit2))
  mu.IPW <- function(alpha){
    mean( ((policy(alpha,A.unit1,A.unit2)*Y.unit1 +
              policy(alpha,A.unit2,A.unit1)*Y.unit2))/2/CPS.Est )
  }
  return(sapply(alpha.grid,function(v){mu.IPW(v)}-mu.IPW(0)))
}

B <- 100; OE.IPW.Boot.Matrix <- matrix(0,11,B)
for(bb in 1:B){
  Cl.Index <- sample(M,M,replace=TRUE)
  OE.IPW.Boot.Matrix[,bb] <-
    OE.IPW.Boot(Data[as.vector(rbind(2*Cl.Index-1,2*Cl.Index)),])
}
OE.CI <- apply(OE.IPW.Boot.Matrix,1,function(v){quantile(v,c(0.025,0.975))})
```



Black $= \widehat{\mathrm{OE}}(\alpha', \alpha = 0)$; Red $= 95\%$ CI

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Cluster Randomized Trial

- **CRT**: an intervention is randomized at the cluster level

- CRTs might be preferred over individual-level randomized controlled trials in the following situations:
  - When the target of the intervention is a broader entity (e.g., hospital, school, community) rather than an individual (e.g., patient, student)
  - When there is a significant potential for interference or contamination between individuals within the same cluster
  - When interventions are naturally delivered at the group level (e.g., policies, educational programs, public health campaigns)
  - When logistical, ethical, or practical considerations make individual randomization difficult or impossible
  - When measuring outcomes that are inherently collective (e.g., infection rates, school performance, environmental exposure)
  - When the goal is to evaluate both individual- and cluster-level effects simultaneously

Examples: Cluster Randomized Trial

- Development economics (Abhijit Banerjee, Esther Duflo, Michael Kremer; 2019 econ Nobel prize)
  - Duflo et al. (2015): 328 schools in Kenya were randomly assigned to one of four groups:
    (i) control, (ii) a subsidy program, (iii) an HIV education program, or (iv) both interventions
    Students' outcomes (attendance rate, early marriage, early pregnancy) were subsequently measured

- Political science
  - Gerber and Green (2000): The get-out-the-vote intervention was assigned at the household level
    Individuals' voter turnout was subsequently measured

- Public health
  - Cowling et al. (2009): 407 households were randomly assigned to one of three groups:
    (i) control, (ii) free hand hygiene; (iii) free face masks
    Individuals' flu infection status were subsequently measured

Estimands & Inference

- **Recall**: Under partial interference,

$$Y_{ij}(a_{ij}, \boldsymbol{a}_{i(-j)}) = \text{potential outcome of unit } ij \text{ if } \{A_{ij} = a_{ij}, \boldsymbol{A}_{i(-j)} = \boldsymbol{a}_{i(-j)}\}$$

- Under CRT, we always have $\boldsymbol{a}_i = (1, \ldots, 1)^\intercal$ or $\boldsymbol{a}_i = (0, \ldots, 0)^\intercal$

- Thus, even though partial interference is possible, the relevant potential outcomes simplify to

$$Y_{ij}(a) = \text{potential outcome of unit } ij \text{ when cluster } i \text{ receives treatment level } a$$

- Therefore, for each unit, the only contrast of interest is $Y_{ij}(1) - Y_{ij}(0)$

- From a linear model perspective: Estimate τ_A:

$$Y_{ij} = \beta_0 + \tau_A A_i + \beta_X^\intercal X_{ij} + \epsilon_{ij}$$

- Wang et al. (2024): Influence function-based inference for $\mu(1) - \mu(0)$ where $\mu(a)$ is defined either by:

$$\text{Cluster-average}: \quad \mu(a) = \mathrm{E}\left\{ \frac{\sum_{j=1}^{N_i} Y_{ij}(a)}{N_i} \right\} \qquad \text{Individual-average}: \quad \mu(a) = \frac{\mathrm{E}\{\sum_{j=1}^{N_i} Y_{ij}(a)\}}{\mathrm{E}(N_i)}$$

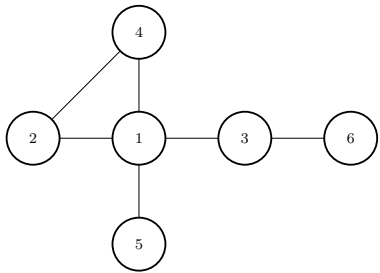
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Network Data

- Denote undirected network as a pair $(\mathcal{N}, \mathcal{E})$ = (Node, Edge) with $|\mathcal{N}| = N$

$$\mathcal{N} = \{1, 2, 3, 4, 5, 6\} \ , \quad \mathcal{E} = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$



- $\mathcal{N}_i$ = 1st-order neighbors of unit i with $|\mathcal{N}_i| = N_i$
  - $\mathcal{N}_2 = \{1, 4\}$, $\mathcal{N}_6 = \{3\}$

Neighborhood Interference

- **Recall**: too many potential outcomes under interference
- Need to reduce the number of potential outcomes for identification

Assumption (Neighborhood Interference (Forastiere et al., 2021))

$Y_i(\boldsymbol{a}) = Y_i(a_i, g(\boldsymbol{a}_{\mathcal{N}_i}))$ *for some* g_i

Assumption (Consistency)

$Y_i = Y_i(A_i, G_i)$ *where* $G_i = g_i(\boldsymbol{A}_{\mathcal{N}_i})$

- g_i is a kind of the exposure mapping; some examples:

$$g_i(\boldsymbol{a}_{\mathcal{N}_i}) = \sum_{j \in \mathcal{N}_i} a_{ij} = \text{total number of treated neighbors}$$

$$g_i(\boldsymbol{a}_{\mathcal{N}_i}) = \frac{\sum_{j \in \mathcal{N}_i} a_{ij}}{N_i} = \text{proportion of treated neighbors}$$

Estimands Under Neighborhood Interference

- Average dose-response function:

$$\mu(a, g) = \mathrm{E}\big[Y_i(a, g)\big]$$

- Main effect at g:

$$\tau(g) = \mu(1, g) - \mu(0, g)$$

- Spillover effect at a:

$$\delta(a, g) = \mu(a, g) - \mu(a, 0)$$

- Overall main effect:

$$\mathcal{T} = \sum_g \tau(g) \Pr(G_i = g)$$

- Overall spillover effect at a:

$$\Delta(a) = \sum_g \delta(a, g) \Pr(G_i = g)$$

Inference Under Neighborhood Interference

Assumption (Conditional Ignorability)

$Y_i(a,g) \perp\!\!\!\perp (A_i, G_i) \,|\, X_i,\ \forall (a,g)$

Assumption (Positivity)

$\Pr(A_i = a, G_i = g | X_i) > 0,\ \forall (a,g)$

- Identification

$$\begin{aligned}\mu(a,g) &= \mathrm{E}_X\big[\mathrm{E}(Y_i | A_i = a, G_i = g, X_i)\big] \\ &= \mathrm{E}\left[\frac{\mathbb{1}(A_i = a, G_i = g) Y_i}{\Pr(A_i = a, G_i = g | X_i)}\right]\end{aligned}$$

- IPW/OR estimators can be constructed

- A doubly robust estimator can be constructed (Chen et al., 2024; Liu et al., 2025)

Finite vs. Infinite Population Framework

- **Finite Population Framework**:
  The potential outcome $Y_i(\boldsymbol{a})$, covariates $\boldsymbol{X}$, and the interference graph $(\mathcal{N}, \mathcal{E})$ are fixed
  Randomness arises solely from the treatment assignment $\boldsymbol{A} \sim P_A$
- In practice, we often have only one network available, which is the *de facto* population
  Since our focus is on this specific network, generalization beyond it is unnecessary

- **Super Population Framework**:
  The potential outcome $Y_i(\boldsymbol{a})$, the treatment assignment $\boldsymbol{A}$, covariates $\boldsymbol{X}$, and the interference graph $(\mathcal{N}, \mathcal{E})$ are random, and generated from a super population:

$$\{Y_i(\boldsymbol{a}), \boldsymbol{A}, \boldsymbol{X}, \mathcal{N}, \mathcal{E}\} \sim P$$

- The super population framework helps generalize causal conclusions, and it provides a more natural setting for applying statistical methods (e.g., regression, ML)

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Additional Literature: Partial Interference/CRT

- Imai et al. (2021); Park and Kang (2023): Noncompliance under PI/CRTs; effects among NT, CO, AT
  - Imai et al. (2021) assume no spillover effects on noncompliers, which allows for point identification
  - Park and Kang (2023) allow spillover effects on noncompliers, making point identification impossible
- Park et al. (2024); Zhang and Imai (2023): Optimal treatment regime under partial interference
- Papadogeorgou et al. (2019); Lee et al. (2025): Other types of causal effects defined besides the α-B policy

Additional Literature: Network Interference

- Ogburn et al. (2024): Causal structural equation models for a network
- Tchetgen Tchetgen et al. (2021): g-formula for a network accounting for autoregressive structure, say $y_i|y_{-i}$
- Li and Wager (2022): Network interference; network is generated from a graphon model; large-sample asymptotics for treatment effect estimation
- Sävje (2023): Impact of misspecified exposure mapping under network interference
- Laber et al. (2018); Viviano (2024): Optimal treatment regime under network interference
- Egami and Tchetgen Tchetgen (2023): Proximal causal interference under network interference
- Leung (2022): Approximate neighborhood interference: higher-order neighbors may influence the ego's outcome, but their effect diminishes with increasing order

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Assignment

- We analyze the study by York (2019)
  - First, visit https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/MHQXTF and download "Original Format Zip"
  - Please run the following code to obtain the data for analysis

```
library(readstata13)
OriginalData <- read.dta13("Public_R3_Twins17.dta")
N <- nrow(OriginalData)
Data <- data.frame( ClusterID=1:N,
                    UnitID=rep(1:2,each=N),
                    X1=as.numeric(c(OriginalData$zyg,OriginalData$zyg)==1),
                    X2=as.numeric(c(OriginalData$sex1,OriginalData$sex2)==1),
                    X3=c(OriginalData$age1,OriginalData$age2),
                    X4=as.numeric(c(OriginalData$education1,OriginalData$education2)>=4),
                    A=as.numeric(c(OriginalData$condition1,OriginalData$condition2)=="Free Speech"),
                    Y=as.numeric(c(OriginalData$rally1,OriginalData$rally2)>=4) )
Data <- Data[order(Data$ClusterID),]
```

Assignment

- The dataset looks like:

	ClusterID	UnitID	X1	X2	X3	X4	A	Y
1	1	1	1	1	85	1	1	0
168	1	2	1	1	85	1	1	1
2	2	1	1	0	20	0	0	0
169	2	2	1	0	20	0	0	0
3	3	1	1	0	46	0	0	0
170	3	2	1	0	46	0	1	0

Covariate	X_1	$X_1 = 1$ if the twin pair is monozygotic (identical) $X_1 = 0$ if the twin pair is dizygotic (non-identical)
	X_2	$X_2 = 1$ if a person is male $X_2 = 0$ otherwise
	X_3	Age
	X_4	$X_4 = 1$ if a person has a college degree or higher $X_4 = 0$ othwerwise
Treatment	A	$A = 1$ if a person read a news article supporting the KKK $A = 0$ otherwise
Outcome	Y	$Y = 1$ if a person advocates for the KKK's rallies $Y = 0$ otherwise

- Goal: Estimate the causal effect of the treatment on a person's likelihood of advocating for the KKK
- Under partial interference (using each twin pair as a cluster), estimate the overall effect, direct effect, and indirect effect based on the IPW and OR estimators
  - Since the treatment is randomized, the propensity score can be estimated without covariates, e.g., `glm(Y ~ 1)`.
  - Make sure you consider multiple the policy parameter $\alpha, \alpha' \in [0, 1]$
- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

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